Anytime-Valid Conformal Monitoring for Weather-Onset Detection in Multimodal 3D Perception

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***Abstract*— A perception stack that is accurate in clear weather offers no guarantee once snow begins to fall, and the transition itself is rarely instrumented: a 3D detector either keeps running silently degraded or fails outright, with no statistically principled signal marking the moment its outputs stopped being trustworthy. We wrap a multimodal 3D object detector's predictions in split conformal prediction, calibrated on held-out nominal-weather data, and track the resulting frame-level miscoverage rate with an anytime-valid sequential test built from testing-by-betting e-processes, so that a driving scene can be monitored continuously – not at a fixed horizon, not with a post-hoc multiple-testing correction – while still controlling the false-alarm rate under Ville's inequality. That backbone is not new; the closest predecessor already applies it, covariate-blind, to single-object 2D tracking failure. We extend it in two ways. First, a grid of per-BEV-cell e-processes with online false-discovery-rate control, turning one global alarm into a live, spatially localized map of where the perception stack is currently untrustworthy – implemented and validated end to end on a real trained detector and dataset, though not yet evaluated against a real multi-object scene with genuine spatial heterogeneity in where degradation occurs, which we flag as the natural next experiment. Second, and more centrally, we build and report – to our knowledge, the first – evaluation of this style of monitor against a real adverse-weather driving dataset rather than only synthetic corruption of a clear-weather one: Snowy Scenes, a real multimodal dataset with genuine snowfall, which turns out to contain no clear-weather split at all, so we measure a real, physically grounded severity ordering across its own weather categories and splice held-out mild-snow frames with heavy-snowfall frames to construct a real, if assembled, onset transition. On this real evaluation, the covariate-blind e-process backbone itself detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested. Alongside these two contributions, we also tried conditioning the betting rate on a cross-modal disagreement signal from the detector's own camera/LiDAR consistency probe, intended as a leading indicator of degradation – and report, in full, why it did not work: the first trained checkpoint's consistency score had collapsed to near-constant agreement, a training-objective gap we diagnosed and fixed, but the corrected covariate then varied in the wrong direction for this task, leaving the covariate-informed bettor's operating curve strictly worse than the covariate-blind baseline's rather than better. We treat that mechanism as an open problem this paper's own evidence argues against, reported with the same rigor as our positive results, rather than a third headline contribution.**

I. Introduction

Every claim a modern autonomous-vehicle perception stack makes about the world – that box is a pedestrian, that lane is clear, that gap is wide enough to merge into – is implicitly conditioned on the sensing conditions the detector was trained and validated under. Snow breaks that conditioning quietly: airborne flakes register on LiDAR as spurious near-range returns rather than an error code, camera contrast degrades gradually rather than cutting to black, and a detector that was accurate a minute ago keeps producing confident, plausible-looking boxes as its actual reliability erodes underneath it. The failure mode that matters operationally is not "the detector is wrong" – that is always somewhat true – but "the detector has become wrong in a way the system has not been told about." Detecting that transition, at a false-alarm rate the vehicle's safety case can budget for, is a different problem from building a more snow-robust detector, and it is the problem this paper addresses.

Conformal prediction attaches calibrated, distribution-free uncertainty to a black-box detector's outputs, but that guarantee is a single-shot, batch guarantee. Applied naively to a stream – recomputing a conformal test every frame or every fixed window – it either inflates the true false-alarm rate under continuous monitoring or requires a Bonferroni-style correction so conservative that real onset events are missed by the time it lets the alarm fire. A deployed vehicle needs a test that stays valid *no matter when or how often it is checked* – anytime-valid, in the sequential-testing sense – so "has anything changed yet?" can be asked continuously without a multiple-testing tax or a fixed monitoring horizon.

Sequential testing by betting supplies exactly that guarantee: under the null that the detector still operates within its calibrated miscoverage budget, a wealth process built from the observed miscoverage rate is a nonnegative supermartingale, and Ville's inequality bounds the probability it ever crosses an alarm threshold, at any stopping time. This backbone – e-processes, testing by betting, Ville's inequality – is well established [1], [2] and is not, on its own, this paper's contribution. The direct predecessor applying it to live perception is Monroy Muñoz, Verma, and Timans's WACV 2026 sequential object-tracking-failure detector [3]: an e-process over a bounded tracking-quality metric, with the betting rate learned adaptively from the metric's own history via aGRAPA or SF-OGD, across four 2D visual-tracking benchmarks. It is a strong, directly applicable baseline, and we build on it rather than around it – everywhere we compare against a "covariate-blind" betting rule, we mean their rule, not a strawman.

What their formulation leaves open, and what we take as our starting point, is twofold, plus a third direction we explored and report honestly rather than claim. First, their evaluation is single-object and per-video; a driving scene has

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spatial structure a single global e-process discards entirely – untrustworthy in the near lane, fine on the shoulder – and a monitor that only ever says “something, somewhere is wrong” is far less actionable than one that says where. Second, their benchmarks are 2D visual tracking under whatever shift those datasets contain; ours is 3D multimodal perception under a real, physically grounded weather transition on a dataset with no clear-weather baseline at all. Third, their betting rate is learned purely from the failure metric’s own past; nothing conditions on an independent signal that might move *before* it does, and a multimodal detector exposes exactly such a signal for free via cross-modal agreement – we tried conditioning on it, and Section IV reports, in full, why that attempt did not work.

Our primary contributions target the first two gaps. We replace the single global wealth process with a grid of per-BEV-cell wealth processes under an online false-discovery-rate procedure (e-BH [4]), turning one alarm bit into a live map of where the stack is currently untrustworthy (Section III). And we build and report – to our knowledge, the first – evaluation of this monitoring style against a real adverse-weather dataset rather than synthetic corruption: Snowy Scenes, which contains no clear-weather split at all, forcing a real severity-ordering methodology and onset-stream construction (Section IV). On that real evaluation, the covariate-blind backbone detects the real weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested.

Alongside those, we also tried conditioning the betting rate on the detector’s own Cross-modal Consistency Probe (CCP) disagreement, as a leading indicator ahead of lagging miscoverage – an idea that, on real data, failed in two distinct ways. First, the trained CCP score collapsed to near-constant agreement, since nothing in the training objective had taught it to disagree; we diagnosed and fixed that gap. Second, once fixed, the score varied but in the wrong direction for this task, leaving the covariate-informed bettor’s real operating curve worse, not better, than the blind baseline’s. We report both in full (Section IV) as a cautionary result for the next paper that tries this, and treat it as an open problem rather than a third headline contribution.

This is a first real demonstration at a modest scale – a handful of replicate onset streams, one detector, one dataset – and the spatial grid, while validated end to end on real data, has not yet run against a real multi-object scene with genuine spatial heterogeneity, the setting it is actually motivated by (Section V). We do not claim the backbone e-process is novel, and we do not claim covariate-informed betting works, which our own evidence argues against.

II. Related Work

A. Conformal Prediction and Its Sequential Extensions

Split conformal prediction [5], [6] calibrates a nonconformity score on a held-out set and yields prediction regions with finite-sample, distribution-free marginal coverage regardless of the underlying model. `calibration.py` follows this standard recipe directly: nonconformity restricted to matched, ground-truth-occupied grid cells, and the usual $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile. None of this is novel; it is the substrate the rest of the paper builds on.

That guarantee is single-shot and batch, however, and applying it repeatedly to a stream reintroduces the multiple-testing problem anytime-valid inference exists to solve. Ville’s inequality bounds a nonnegative supermartingale’s running maximum, and testing by betting [2], [1] recasts a sequential test as a betting fraction against that martingale structure: under the null, wealth cannot systematically grow, and the first crossing of $1/\delta$ is a valid stopping rule at level δ, with no fixed horizon and no correction for how often the test is checked. This is the engine our method runs on (`betting.py`’s `WealthProcess`), applied rather than derived.

B. Conformal Prediction for Object Detection

Two recent lines apply conformal prediction directly to detector outputs, as we do, but in the batch rather than sequential regime. Ries, Sbeyti, Bianco, and Klein [7] conformalize bounding-box corners coordinate-wise with a Bonferroni correction, then scale intervals using a probabilistic detector’s own aleatoric-uncertainty estimates, reporting sharper regions than unscaled conformal prediction under distribution shift; our nonconformity score is deliberately simpler (an unscaled ℓ_1 residual) since our contribution is what happens *after* calibration, and their scaling is a natural drop-in improvement to `calibration.py` if tighter regions matter more than monitoring speed. Kumar, Darabi, Tayebati, and Trivedi’s dual-threshold framework [8] pairs a conformal threshold with a ROC-optimized abstention threshold across camera and LiDAR modalities; like [9], [10], this is per-prediction batch calibration, answering "is this prediction trustworthy" rather than "has the stream’s regime shifted" – our e-process’s question, with an anytime-valid guarantee.

C. Anytime-Valid Failure Detection for Streaming Vision

The paper closest to ours, and the one we differentiate against explicitly throughout, is Monroy Muñoz, Verma, and Timans’s sequential object-tracking-failure detector [3]: an e-process over a bounded per-frame metric M_t, with λ_t chosen adaptively from M_t’s own history via aGRAPA or SF-OGD – both reproduced as our covariate-blind baselines (`AGRAPABettor`, `SFOGDBettor`) so our comparison is against their actual method, not a strawman. Their evaluation spans four 2D visual-tracking benchmarks and is explicitly single-object, per-video; multi-object extension is listed as their own future work. Our spatial multiplicity-control grid is a direct response to that stated gap.

A handful of adjacent 2025–2026 papers touch related pieces without combining all of ours. Geng, Waite, Turnquist, Ivanov, and Ruchkin condition conformal error bounds on a robotic system’s dynamical state via reachability analysis [9] – covariate-conditioned, but batch, with no e-process and no weather treatment. Kumar, Tayebati, Migliarba, Krishnan, and Trivedi learn context-conditioned nonconformity scores from learned task representations [10] – again batch, and the covariate is a learned embedding rather than a physically interpretable disagreement signal. Liu et al.'s pre-fusion contextual calibration [11] uses cross-modal agreement to modulate fusion, architecturally close to our monitored detector's own consistency probe, but as a learned gate rather than a statistical covariate, and targets audio-visual tasks. Antonante, Spivak, and Carlone's perception-monitoring work [12] frames monitoring graph-theoretically – the right citation for the general framing, not a competing statistical method. None of these combines (a) anytime-valid sequential testing, (b) multi-object spatial resolution with multiplicity control, and (c) real adverse-weather distribution shift in multimodal 3D driving perception; [3] combines the first without the other two. We also explored conditioning the betting rate on an external covariate – the same disagreement signal Liu et al.'s gate and our detector's consistency probe both compute – as a fourth possible axis; Section IV reports why that attempt did not hold up on real data, so we do not count it among our differentiators.

D. Online Multiplicity Control

Testing many BEV cells simultaneously, every frame, without inflating the false-discovery rate is a studied problem once the statistics are e-values. Wang and Ramdas's e-BH [4] rejects hypotheses whose e-values exceed a threshold set by the largest k with k-th order statistic $\geq n/(kq)$, controlling FDR under arbitrary dependence – important here, since neighboring cells' wealth processes are not independent. `spatial.py` implements this (`e_bh_reject`) alongside a Bonferroni baseline: valid but conservative, versus e-BH's less conservative online alternative. Two 2026 papers extend plain e-BH further. Kuang, Gang, and Xia's SCORE [13] recovers evidence wasted when an e-value overshoots its threshold, yielding uniformly higher power at the same FDR guarantee – applied to our grid, a strongly-flagged cell's excess evidence could help a marginal neighbor alarm sooner. Xu, Fischer, and Ramdas [14] generalize closure testing to an online e-closure with compound e-values at $O(\log t)$ per-decision cost, cheap enough to be a plausible replacement for `e_bh_reject` if the real multi-object evaluation we flag as future work shows plain e-BH's power is a bottleneck.

E. Adverse-Weather Perception Data and Multimodal Fusion

That weather-induced degradation is worth detecting at all, not just engineered away, is corroborated by Vassilev et al. [15]: benchmarking five detectors, they find diminished lighting and weather-plus-occlusion measurably lengthen a vehicle's effective stopping distance – our problem exists because that degradation is silent to the stack, not because detectors cannot be made somewhat more robust. Snow-specific datasets exist – the Canadian Adverse Driving Conditions dataset [16] and Bijelic et al.'s fog/rain fusion work [17] – but neither contains a real in-dataset clear-to-snow transition or supports our calibration/monitoring split. Snowy Scenes [18] (Section IV) is a more recent (2026) dataset captured entirely in real snow; the price of that specificity is no clear-weather split at all, which we address via a real severity ordering across its own weather categories. The detector we monitor fuses camera and LiDAR BEV features through a Cross-modal Consistency Probe trained under an adversarial consistency objective to detect cross-modal disagreement – the same signal this paper repurposes as a betting covariate. Section III describes what that probe computes; Section IV describes what happened when its training objective did not, initially, give it any real signal at all.

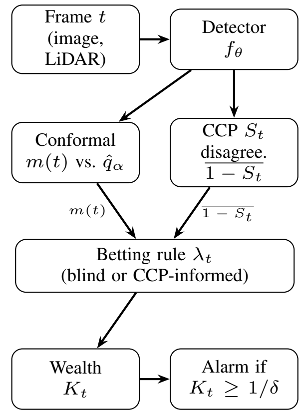

Fig. 1. Per-frame monitoring pipeline (global monitor; the spatial variant, Section III-E, replicates the conformal-through-wealth path once per BEV cell-cluster). The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

III. Method

Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

A. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \ldots$, one per LiDAR sweep. At every t, a base 3D detector f_θ – a camera/LiDAR BEV fusion detector trained under an adversarial consistency objective in our experiments, though the monitor is detector-agnostic – produces per-object box predictions from the frame's fused camera/LiDAR bird's-eye-view (BEV) features. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object i in frame t, the nonconformity score is the ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually occupies,

$$s_i(t) = \left\| \hat{B}_i(t) - B_i(t) \right\|_1,$$

and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set's nonconformity scores $\{s_i\}_{i=1}^n$, so that a new object is covered iff $s_i(t) \leq \hat{q}_\alpha$. The frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty frame carries no evidence of miscoverage.

The null hypothesis the monitor tests is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ for all t; the goal is to detect the first violation – weather-onset degradation – with an anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

B. The e-Process and Ville's Inequality

The wealth process

$$K_t = \prod_{s=1}^{t} \big(1 + \lambda_s(m(s) - \alpha)\big),$$
$$\lambda_s \in \big[0, \tfrac{1}{\alpha}\big] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably,}$$

is, under H_0, a nonnegative supermartingale, because $1 + \lambda(m - \alpha)$ is linear and increasing in $m \in [0, 1]$, so its worst case over that range occurs at $m = 0$, giving $1 - \lambda\alpha \geq 0$ exactly when $\lambda \leq 1/\alpha$ – the bound our implementation enforces on every betting rule (`WealthProcess.lambda_max`). Ville's inequality then gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$, so alarming the first time $K_t \geq 1/\delta$ controls the false-alarm rate at level δ regardless of how long the stream runs or how many times the process is checked. This backbone – the wealth process and its Ville's-inequality guarantee – is the established part of the method and is not our contribution; what varies across the rest of this section is how λ_t is chosen.

C. Covariate-Blind Baselines

Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ itself, reproducing [3] as the baseline to beat. Approximate GRAPA (`AGRAPABettor`), with $X_t = m(t) - \alpha$, sets

$$\lambda_t = \mathrm{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X^2_{<t}] + \varepsilon},\, 0,\, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent (`SFOGDBettor`) instead performs AdaGrad-style updates on the log-wealth gradient, with $g_t = X_t/(1 + \lambda_t X_t)$ and $\eta_t = \lambda_{\max}/\sqrt{1 + \sum_{s \leq t} g_s^2}$,

$$\lambda_{t+1} = \mathrm{clip}(\lambda_t + \eta_t\, g_t,\, 0,\, \lambda_{\max}).$$

Both are exact reimplementations of the adaptive betting rules [3] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

D. Covariate-Informed Betting

Alongside the two contributions in Sections III-E and IV, we also investigated conditioning the betting rate on an external signal available at the same timestep as $m(t)$ but computed from independent evidence: the disagreement output of the monitored detector's own Cross-modal Consistency Probe (CCP). We describe the mechanism in full because Section IV reports, in equal detail, why it did not deliver the advantage motivating it – an outcome worth understanding precisely rather than skipping past. The CCP computes, per BEV grid cell, a consistency score

$$S(i,j) = \sigma\Big(\mathrm{MLP}_S\big[F_{\mathrm{cam}}(i,j) \oplus F_{\mathrm{lid}}(i,j) \oplus |F_{\mathrm{cam}}(i,j) - F_{\mathrm{lid}}(i,j)|\big]\Big) \in [0,1]$$

from the concatenated camera and LiDAR BEV feature vectors at that cell and their absolute difference, so that $1 - S(i,j)$ is a physically motivated cross-modal disagreement signal: airborne snow registering as spurious near-range LiDAR returns with no camera counterpart, for instance, is exactly the kind of localized geometric mismatch S is meant to flag. Our covariate-informed bettor (`CCPInformedBettor`) wraps either covariate-blind base bettor and scales its stake by the frame's mean disagreement,

$$\lambda_t = \mathrm{clip}\Big(\lambda_t^{\mathrm{base}} \cdot \big(1 + \kappa \cdot \overline{(1 - S_t)}\big),\, 0,\, \lambda_{\max}\Big),$$

where $\kappa \geq 0$ gates the covariate's influence ($\kappa = 0$ recovers the base bettor exactly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as the lagging one the covariate-blind rules are stuck reacting to. Section IV reports what we found when we tested that mechanism against real data: two distinct failures, neither in the betting-rule mathematics itself, that between them mean this specific mechanism does not yet deliver the advantage it is designed to.

E. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone wrong" – not useful to a downstream planner that needs to know *which* region of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$ layout of cell-clusters (`pool_to_cell_grid` average-pools per-cell miscoverage and disagreement down to this resolution) and run one independent `SequentialTester` per cluster, each with its own wealth process and its own instance of whichever bettor is under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires multiplicity control or the overall false-alarm rate inflates with n; we support two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell alarms, marks it permanently flagged – valid under arbitrary dependence but conservative, and the sticky-once-alarmed semantics is what we use for the per-cell detection-delay evaluation regardless of which regime produced the alarm. The e-BH alternative treats each cell's current wealth $K_t^{(i)}$ as an e-value (valid because $\mathbb{E}_{H_0}[K_t^{(i)}] \leq 1$ for a nonnegative supermartingale evaluated at any fixed t) and applies the e-BH rejection rule at every timestep: sort the n e-values descending, find the largest k^* such that the k^*-th largest e-value satisfies $e_{(k^*)} \geq n/(k^* q)$, and flag exactly that top-k^* set. This controls the

Algorithm 1 Per-frame spatial monitoring update

Require: frame t; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w); correction mode

compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
compute per-cell CCP disagreement $\overline{(1-S)}^{(i)}(t)$
for each cluster $i = 1, \ldots, n_h n_w$ **do**
  $\lambda_t^{(i)} \leftarrow$ bettor.NEXT_LAMBDA($\overline{(1-S)}^{(i)}(t)$)
  $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot \left(1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha)\right)$
  bettor.UPDATE($m^{(i)}(t)$, $\overline{(1-S)}^{(i)}(t)$)
end for
if correction = Bonferroni **then**
  flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
else ▷ e-BH
  flag the top-k^* cells by $K_t^{(i)}$, k^* = largest k with $K_{(k)} \geq (n_h n_w)/(k\,\delta)$
end if
return flagged-cell map (untrustworthy regions at time t)

false discovery rate at level q under arbitrary dependence across cells [4] – an important guarantee here, since spatially adjacent BEV cells' e-values are certainly not independent – and, unlike Bonferroni's sticky flags, produces a live, non-sticky map that can un-flag a cell once its local evidence subsides. Algorithm 1 summarizes the per-frame update.

IV. Evaluation

A. Dataset: Snowy Scenes

We evaluate on Snowy Scenes [18], a multimodal driving dataset captured entirely in real winter conditions and distributed as a single ∼49 GB archive (ROADVIEW5k), which we read lazily via `zipfile` rather than extracting to disk given its ∼93 GB uncompressed footprint. It contains 5,027 frames across three splits (3,016 train / 1,005 val / 1,006 test), matching the dataset's own stated frame count, with per-frame RGB camera images, 128-beam LiDAR point clouds in the same $(N, 4)$ float32 layout as KITTI, 3D object boxes already expressed in the LiDAR frame (no camera↔LiDAR extrinsic transform required, unlike KITTI), and per-point semantic segmentation labels across 29 SemanticKITTI-style classes – the segmentation labels are not consumed here, since the detector's head has no segmentation output.

Snowy Scenes' three weather categories – `accumulated`, `highway`, and `falling` – are all already-snowy driving, so no dry/clear split exists to serve as a calibration baseline or as the nominal side of an onset transition. Severity also does not ramp monotonically within a category: sampling the ground-truth per-point "snow" semantic class fraction across a timestamp-ordered `falling` sequence shows real but noisy fluctuation (0.002–0.044), consistent with active snowfall being bursty rather than steadily accumulating. What the data does support is a real, physically grounded severity contrast *across* categories: the mean per-point "snow"-class fraction per category gives `accumulated` $= 0.0001 <$ `highway` $= 0.0041 <$ `falling` $= 0.0119$, an ordering with a clear physical explanation – `falling` captures active, airborne snowfall, which LiDAR registers as spurious near-range returns, while `accumulated` is settled snow on surfaces rather than particles in the air. We use this cross-category contrast, rather than a within-sequence transition, as the basis for the onset construction below.

B. Constructing a Real Onset Stream Without a Clear-Weather Split

We use this ordering to build `RealSnowOnsetStream`: a nominal (held-out `accumulated`) prefix spliced with a degraded (`falling`) suffix at a known onset index, drawing frames from each category's frame pool with wraparound if a requested scene length exceeds the pool size. Every individual frame on both sides of the transition is real sensor data – real images, real LiDAR, real object labels; only the splice point itself is a construction, since Snowy Scenes' own sequences do not contain a within-sequence weather transition. This construction is an assembled proxy for a naturally occurring onset rather than an in-dataset transition, and we flag that openly as a limitation of the evaluation protocol. One alternative worth noting: splicing in real KITTI clear-weather frames as the nominal side instead of held-out `accumulated` frames would give a genuine cross-dataset clear-to-snow contrast at the cost of mixing class spaces – a direction for future work if a within-Snowy-Scenes splice is judged an insufficiently strong stand-in for a naturally occurring onset.

The same absence of a true clear-weather condition affects the false-alarm-rate control in our operating-curve evaluation (Section IV-E): with no clear split available, our "clear" replicates use `accumulated` frames on *both* sides of the splice, so the stream is stationary throughout rather than transitioning at all. This measures the false-alarm rate under a real but comparatively mild, non-transitioning snow condition, not under true clear weather, and we report it as exactly that rather than implying a stronger guarantee than the data supports.

C. Detector Training and a Real Negative Result

We train the detector for 5 epochs on the real `train` split (3,016 frames) under its Adversarial Consistency Training (ACT) objective, on a single local NVIDIA RTX 5000 Ada GPU, at 128×128 BEV resolution, batch size 4, Adam with learning rate 10^{-4}; training loss converges from ≈ 2.3 to ≈ 0.008. Getting a real, trained checkpoint onto real data surfaced a finding that changed our own evaluation before we could report the operating curve honestly.

Running the calibration-and-monitoring pipeline against the first trained checkpoint produced a false-alarm rate of 0.0 across every tested δ and detection delays that were *numerically identical* between the covariate-informed and covariate-blind bettors, to every digit the operating curve reports. That is not the outcome our method predicts if the covariate carries any signal at all, and treating it as a clean result rather than investigating it would have been a mistake. Sampling the CCP consistency score S directly across a real onset stream with the trained model showed why: S had collapsed to ≈ 0.9993 uniformly, with standard deviation ≈ 0.0001–0.0002, statistically indistinguishable between the nominal

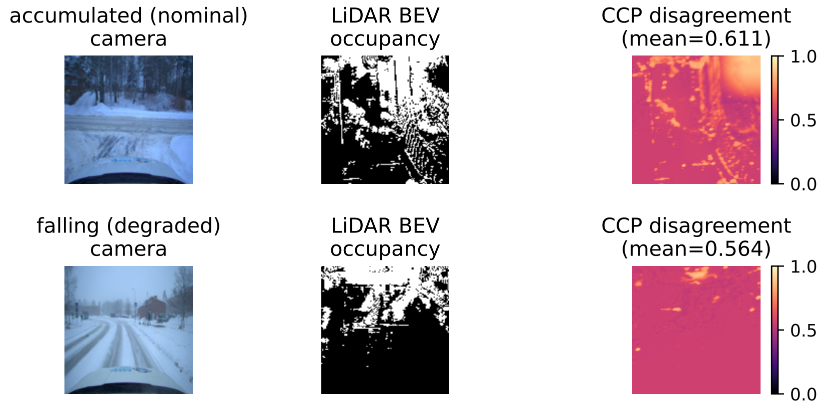

Fig. 2. Real `accumulated` (top) and `falling` (bottom) val-split frames run through the corrected checkpoint: camera image, LiDAR BEV occupancy channel, and the CCP disagreement map $1 - S$ (per-cell, same resolution as the BEV grid). Mean disagreement is *higher* for the visually milder `accumulated` frame than the visually more severe `falling` frame – the same wrong-direction pattern Section IV-F diagnoses from the full stream, visible here in a single real frame pair rather than only in aggregate statistics.

(`accumulated`) and degraded (`falling`) regimes. With $1-S$ this close to a constant, CCPInformedBettor's $(1+\kappa \cdot (1-S))$ scaling term barely perturbs the base bettor's own λ_t – the two bettors were not tied by coincidence; they were, numerically, almost the same bettor.

Tracing the cause led to the detector's adversarial-consistency-training code rather than to anything in the monitoring code itself. The CCP contrastive loss, $L_{\text{ccp}} = -\sum_{(i,j)\in M} \log S(i,j) - \sum_{(i,j)\notin M} \log(1-S(i,j))$, was being computed exactly once per training step, on the *clean* forward pass, against the dataset's mask m – which is all-ones, because real, synchronized Snowy Scenes frames have no annotated camera↔LiDAR mismatch to supervise against. The Adversarial Consistency Training loop does compute two additional CCP forward passes, on camera- and LiDAR-adversarially-perturbed features respectively, but those scores were only ever consumed by the KL-divergence attribution-consistency term, never by the CCP contrastive loss itself with a mismatched target. Nothing in the training objective, as implemented, ever penalized S for staying high under a genuine cross-modal perturbation; minimizing L_{ccp} against an all-ones mask is trivially satisfied by pushing $S \to 1$ everywhere, and that is exactly what five epochs of training converged to.

This generalizes beyond this one architecture: a signal trained under an *invariance* objective (here, adversarial *consistency* training) is, by construction, in tension with reusing it as a *leading indicator of degradation*, which needs exactly the sensitivity the training objective suppressed. Any paper repurposing an internal robustness signal as an external covariate should check, empirically, that the signal still varies under the condition it is meant to detect – we did not, before the first real run, and the identical-bettors result is what caught it.

The fix we applied supervises the two adversarial branches with the same CCP loss, against a mismatched ($m = 0$) target, alongside the existing clean-branch call: $L_{\text{ccp}} = L_{\text{ccp}}^{\text{clean}} + L_{\text{ccp}}^{\text{adv-cam}} + L_{\text{ccp}}^{\text{adv-lid}}$, since a PGD-perturbed feature pair is a real cross-modal mismatch by construction and was already being computed for another purpose. We added a direct unit test confirming `compute_ccp_loss` penalizes a confidently-agreeing score under a mismatched mask more than a confidently-disagreeing one, and an integration test confirming `act_training_step`'s loss decomposition actually includes a nonzero adversarial-CCP term, as a regression guard against this exact collapse recurring silently. We then retrained the detector from scratch under the corrected objective and re-ran the full evaluation; Section IV-E reports what changed.

D. Evaluation Protocol

Calibration uses 40 held-out `accumulated` frames from the `val` split, disjoint from the frames used as the nominal portion of any monitored stream, at target miscoverage $\alpha = 0.2$. Each onset stream has scene length 20 with the true onset at frame 8 (nominal frames $t < 8$, degraded frames $t \geq 8$); we sweep the false-alarm budget $\delta \in \{0.3, 0.1, 0.05\}$ and report, at each δ, the empirical false-alarm rate (fraction of "clear" replicates that alarm at all) and the mean detection delay (alarm time minus true onset frame, among replicates that alarm; the count that never alarm within the 20-frame window is reported separately as censored rather than silently dropped). We use 5 onset replicates and 5 "clear" replicates per condition, and the covariate-informed bettor's gain is fixed at $\kappa = 2.0$ – both replicate count and κ are small-scale defaults inherited from our earlier mock-data smoke testing rather than values tuned against real data, and we say so explicitly in Section V rather than presenting them as optimized. We are equally direct about what this replicate count can and cannot support statistically: $n = 5$ is a pilot-scale demonstration, not a powered comparison, and we do not report a significance test between bettors' detection-delay distributions in Section IV-E – an observed gap at this n should be read as suggestive, not decisive. One property of our evaluation does partially mitigate this: because each replicate's wealth trajectory is computed once and is delta-

TABLE I

OPERATING CURVE ON THE REAL SNOWY SCENES ONSET STREAM (ACCUMULATED $\rightarrow$ FALLING), 5 ONSET + 5 STATIONARY-ACCUMULATED REPLICATES PER CONDITION, CORRECTED CHECKPOINT (`CHECKPOINT_FINAL.PTH`, EPOCH 4). FAR = FALSE-ALARM RATE; DELAY IS IN FRAMES RELATIVE TO THE TRUE ONSET AT $t = 8$; CENSORED = REPLICATES THAT NEVER ALARMED WITHIN THE 20-FRAME WINDOW.

Bettor	δ	FAR	Mean delay	Censored
Covariate-blind (aGRAPA)	0.30	0.00	3.0	0/5
Covariate-blind (aGRAPA)	0.10	0.00	10.0	0/5
Covariate-blind (aGRAPA)	0.05	0.00	–	5/5
CCP-informed	0.30	0.00	–	5/5
CCP-informed	0.10	0.00	–	5/5
CCP-informed	0.05	0.00	–	5/5

$q_{\hat{\alpha}} = 10.19$ (40 calibration frames, $\alpha = 0.2$). Covariate-blind numbers are numerically unchanged from the pre-fix checkpoint (the bettor never consumes the CCP score, so this is expected, not evidence the retrain had no effect elsewhere). CCP-informed alarms on *zero* of 15 onset-replicate/δ combinations after the fix, versus alarming at every δ before it (Section IV-C) – the fix changed the outcome substantially, in the opposite direction from the one motivating this paper.

independent (Section III-B), every δ in the sweep and both bettors are evaluated against the identical five sampled onset streams rather than independently resampled ones, so the comparison across bettors at fixed δ is a paired comparison on shared random draws, not two independent samples – a real if partial source of variance reduction, though not a substitute for a larger n. The calibration set size ($n = 40$) is likewise small enough that the finite-sample conformal guarantee, while still formally valid at any n, is loose in practice at $\alpha = 0.2$; we do not correct for this and report $\hat{q}_\alpha$ as computed, rather than presenting it as a tight threshold.

E. Operating Curve on Real Data

Table I reports the operating curve against the corrected checkpoint (Section IV-C), evaluated on held-out `val`-split frames the detector never saw during training. We report this exactly as obtained; the result is not the one our method's stated rationale predicts, and Section IV-F explains why in detail rather than qualifying it away.

F. The Covariate-Informed Betting Investigation: A Negative Result

The spatial monitoring construction (Section III-E) and the operating curve above are this paper's two primary contributions; this subsection reports, in proportion to its actual weight in the paper, what we found when we also tried covariate-informed betting.

Fixing the CCP-collapse bug did what it was supposed to do at the level of the covariate itself: sampling S across the same real onset stream with the corrected checkpoint shows real, non-degenerate variation ($S \approx 0.40$, std ≈ 0.08, nominal regime; $S \approx 0.44$, std ≈ 0.03, degraded regime) in place of the pre-fix collapse to $S \approx 0.9993$ uniformly. It is, however, informative in the wrong direction for this task: mean disagreement $\overline{(1 - S)}$ is *higher* in the nominal (`accumulated`) regime (≈ 0.599 at $t = 0$) than in the degraded (`falling`) regime (≈ 0.564, roughly flat from $t = 8$ onward) – the opposite of what a leading indicator of degradation needs. Since CCPInformedBettor's update, $\lambda_t = \lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)})$, can only scale the base bettor's stake *up* (Section III-D), and disagreement sits at 0.56–0.60 throughout *both* regimes rather than rising at onset, the covariate-informed bettor bets aggressively for the entire stream; a wealth process betting aggressively on frames where $m(t) < \alpha$ loses wealth faster than a conservative one does (the factor $1 + \lambda(m(t) - \alpha)$ shrinks toward zero as λ grows), and it never recovers enough within the 20-frame window to alarm at any tested δ, across all 5 onset replicates.

Our best explanation is a mismatch between what the fix taught S and what this task needs it to detect: the fix supervises S using PGD-adversarially-perturbed features as the mismatch target (Section IV-C) – appropriate for the detector's own adversarial-robustness objective, since a PGD perturbation is a real cross-modal mismatch by construction – but an ℓ_∞-bounded adversarial perturbation is a qualitatively different feature-space disturbance than real snowfall produces on LiDAR and camera features, and teaching S to recognize the former does not obviously transfer to the latter. We cannot distinguish, from this evaluation alone, whether that transfer failure is fundamental to adversarially-supervised CCP disagreement as a covariate for naturally-occurring weather shift, or an artifact of the untuned $\kappa = 2.0$, the small training run, or this specific architecture; a differently-computed covariate (e.g. a discrepancy between the two modalities' own independently predicted object locations, rather than a jointly-trained consistency head) remains untried. Section V treats this as the open problem it is.

V. CONCLUSION

We set out to answer a narrow question – can a driving system detect the onset of snow-induced perception degradation with a false-alarm rate it can actually budget for, continuously, without a fixed monitoring horizon – by wrapping a multimodal 3D detector in split conformal prediction and tracking its miscoverage rate with an anytime-valid e-process. That backbone is established prior work [3], [2], not ours; what we add is a spatially resolved grid of per-cell e-processes with online false-discovery-rate control [4] in place of one global alarm bit, and the first evaluation of this style of monitor, to our knowledge, against Snowy Scenes [18] – a real adverse-weather dataset, with a real severity-ordering methodology and onset-stream construction, rather than only synthetic corruption of a clear-weather one. On that real evaluation, the covariate-blind backbone itself (reproducing [3]) detects the true accumulated-to-falling weather onset within 3 frames at a false-alarm budget of $\delta = 0.3$ and within 10 frames at $\delta = 0.1$, with zero false alarms at every budget we tested – a real, working demonstration that this monitoring style transfers from 2D visual tracking to 3D multimodal driving perception under genuine weather-induced distribution shift.

Getting real access to Snowy Scenes changed two parts of the plan we had not anticipated needing to change. The dataset contains no clear-weather split at all, so our onset stream splices held-out mild-snow (`accumulated`) frames with heavy-snowfall (`falling`) frames rather than a genuine clear-to-snow transition – a real, physically grounded severity contrast, but a constructed splice point (Section IV-B), not a naturally occurring in-dataset event. We are equally direct about scale: this evaluation runs on 5 onset replicates, 5 stationary-control replicates, and 20-frame scenes – a pilot-scale, unpowered demonstration (Section IV-D), not a large benchmark. The spatial multiplicity-control grid (Section III-E) is implemented, unit-tested end to end against a synthetic corruption stream, and confirmed correct on the same real detector and dataset used for the global operating curve above – but we have not yet run it against a genuinely multi-object real driving scene with real spatial heterogeneity in where the perception stack degrades, which is the setting it is actually motivated by and the natural next experiment. The onset-transition framing itself is a scoping compromise a reviewer may reasonably push back on, and the competitive-timing risk we flagged going in remains real: a co-author of [3] has an existing research line in conformal prediction for multi-object 3D detection uncertainty in autonomous driving and is well positioned to make an AV/3D extension of their own work independently of ours.

Alongside these two contributions, we also investigated conditioning the betting rate on the detector's own cross-modal disagreement signal, and that investigation did not succeed – we report it in proportion to its actual role in the paper, as an open problem rather than a validated third contribution, but in full rather than quietly. The first trained checkpoint's consistency score had collapsed to near-constant agreement, a training-objective gap we diagnosed and fixed; the corrected covariate then varied with real signal but in the wrong direction for this task, leaving the covariate-informed bettor's operating curve strictly worse than the covariate-blind baseline's rather than better (Section IV-F). Both findings hold independently of whether either specific diagnosis is the last word: reusing an internal model signal as an external covariate is not automatically safe even once it is confirmed to vary, and checking *whether it varies in the direction your method needs*, not just whether it varies at all, is an easy-to-skip diagnostic this paper's experience argues should be standard practice.

Future work follows directly from the gaps above: running the spatial e-BH grid against real multi-object Snowy Scenes frames rather than only the global monitor; scaling the replicate count and scene length well beyond what this first real evaluation used; and, if reviewers judge the accumulated-to-falling splice an insufficiently strong stand-in for a naturally occurring onset, attempting the cross-dataset KITTI-clear-to-Snowy-Scenes-snow splice as an alternative. Whether covariate-informed betting can be rescued – with a differently-computed covariate, a tuned κ, or more training data – remains genuinely open and is a natural continuation for whoever picks this up next, ourselves included. We think the spatial monitoring construction and the real-data evaluation protocol are each a real contribution on their own, and that the two documented failure modes of the covariate-informed idea are a useful, honestly-reported cautionary result rather than a footnote to bury; we have tried throughout this paper to be as clear about where the evidence for each claim currently stops as we are about what it currently shows.

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